

Jailbreaking defenses

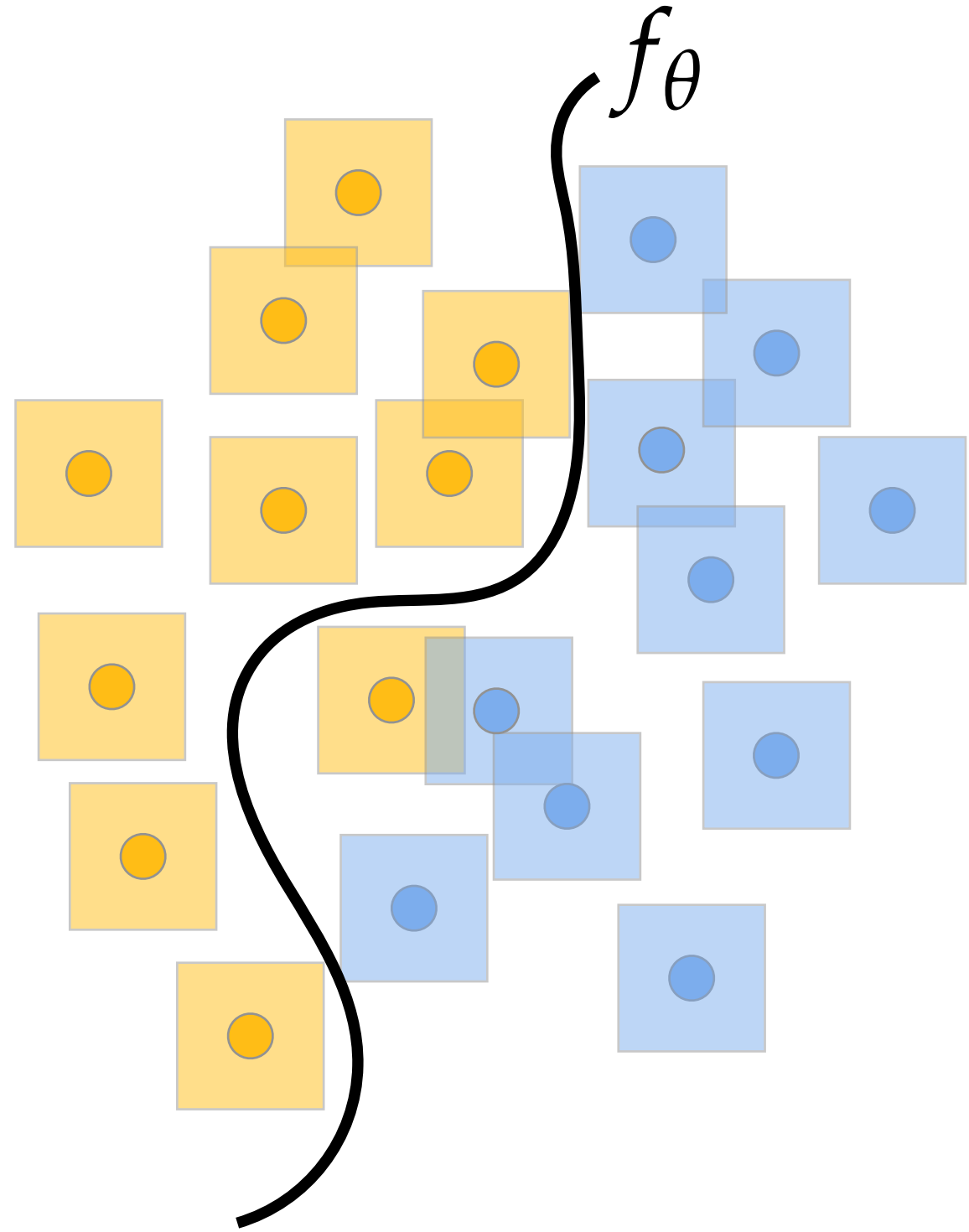
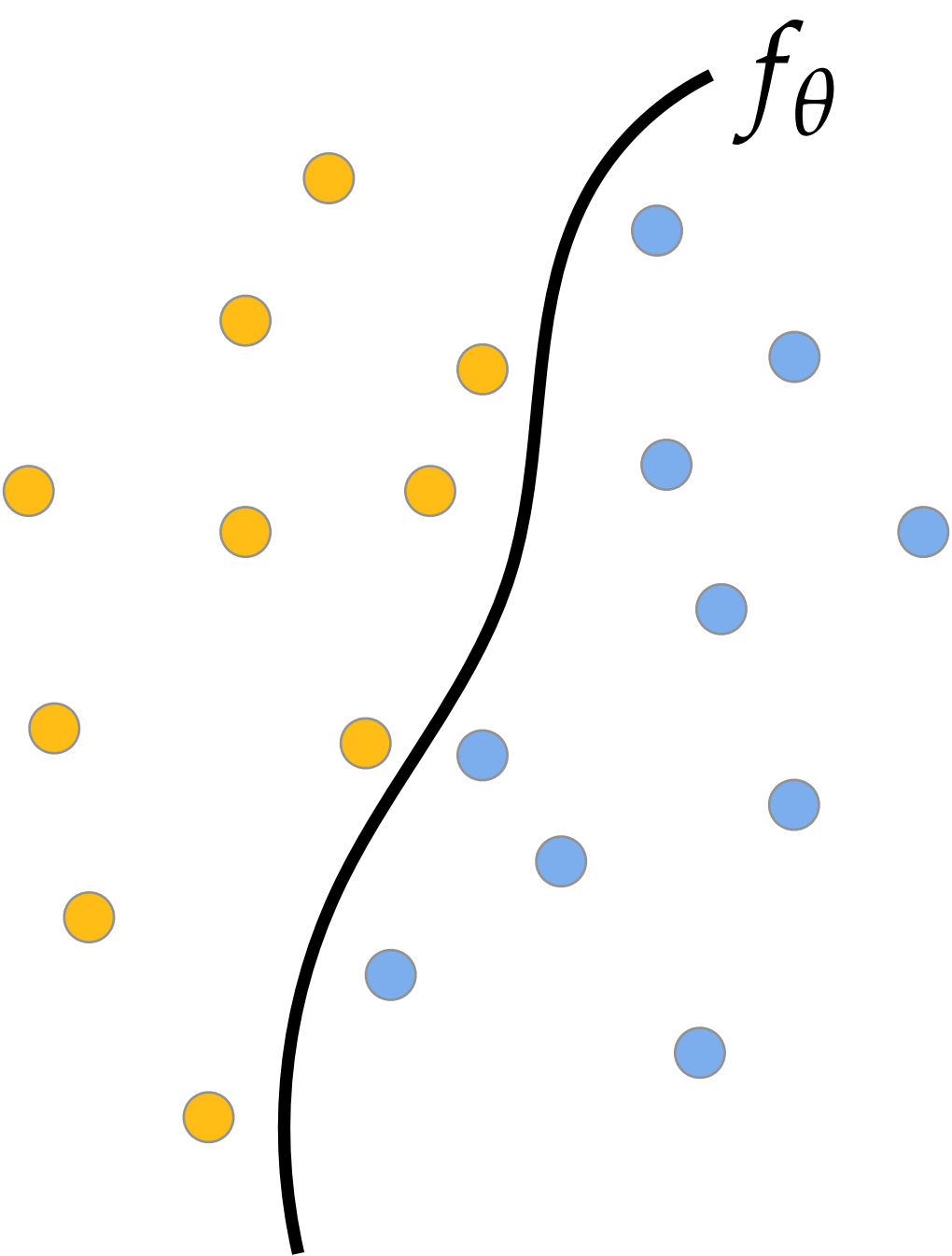
Question: How should we defend against jailbreaking attacks?

1. **Attack mitigation.** Robustness
against adaptive attacks.

2. **Non-conservatism.** Maintain the ability to generate realistic text.

3. *Efficiency. Avoid retraining, maximize query efficiency.*

4. Compatibility. White- & black-box attacks.



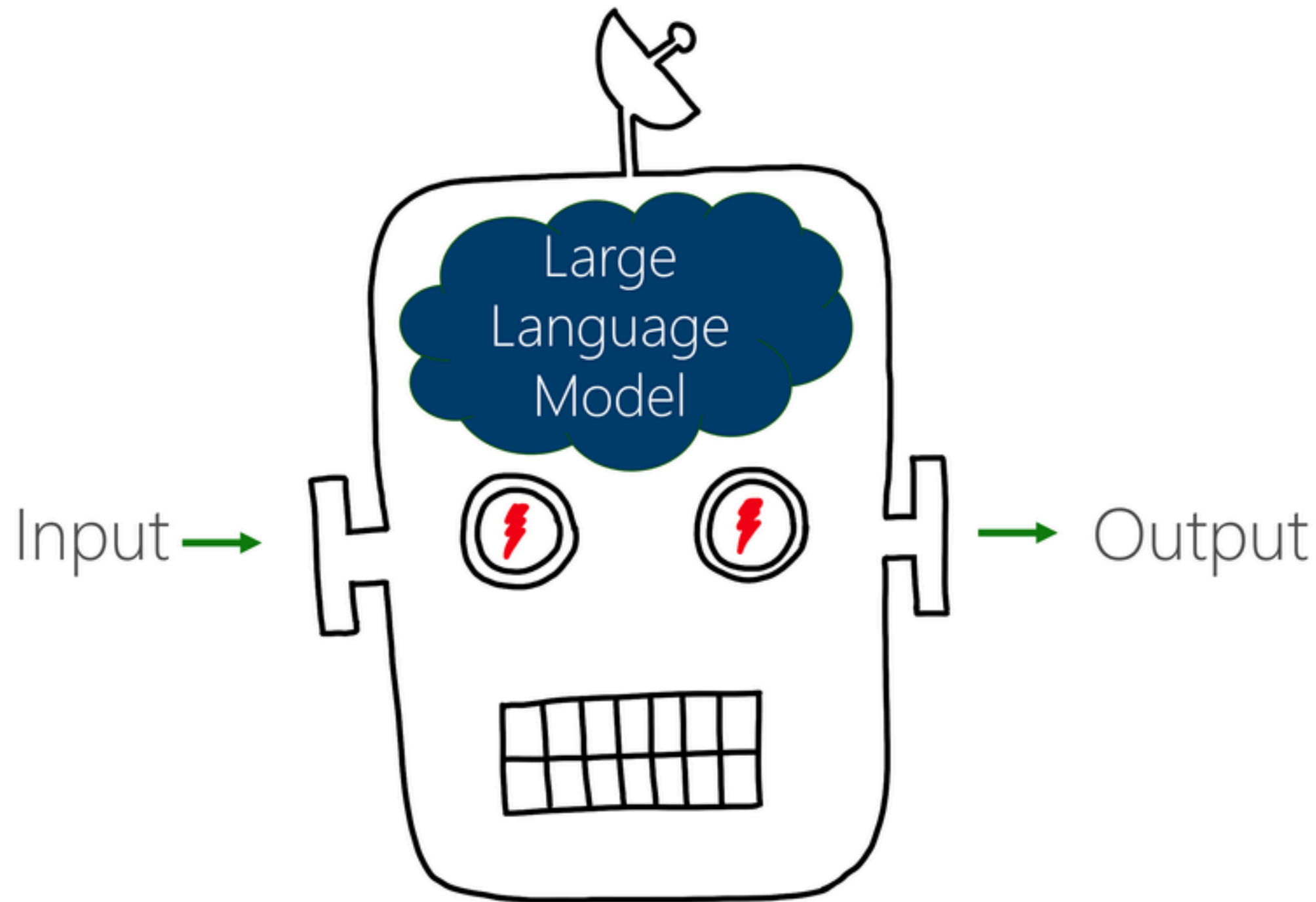


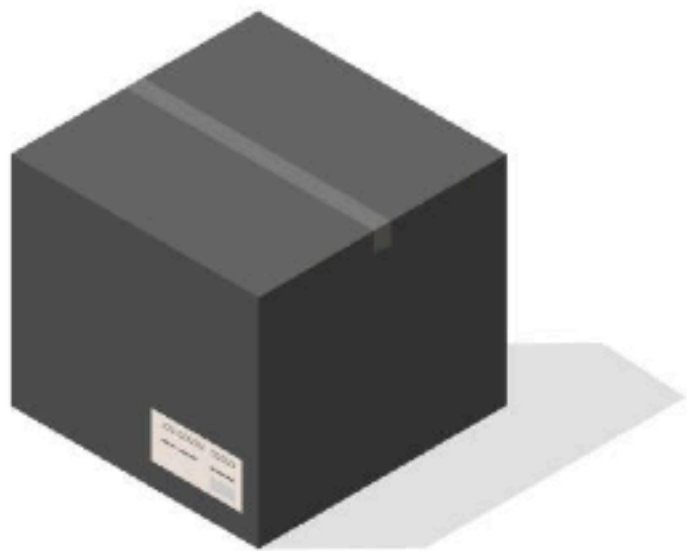
ChatGPT

Gemini



ANTHROPIC
CLAUDE 2





**Black box - we do not
know anything**

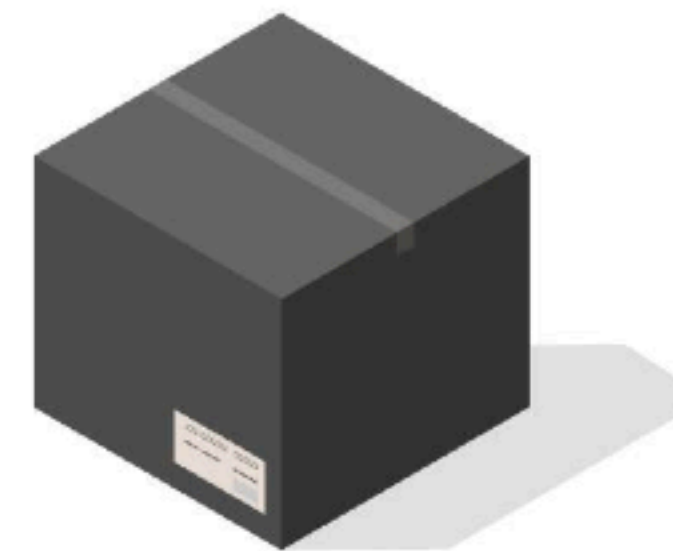


**White box - we know
everything**

Jailbreaking defenses

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1. **Attack mitigation.** Robustness against adaptive attacks.
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3. **Efficiency.** Avoid retraining, maximize query efficiency.
4. **Compatibility.** White- & black-box attacks.



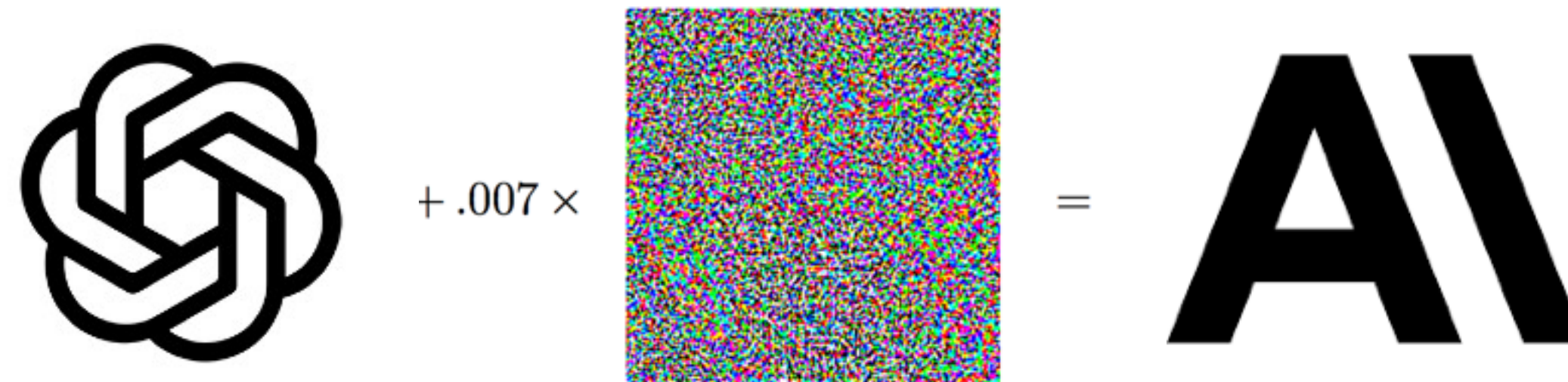
Black box – we do not know anything



White box – we know everything

Jailbreaking defenses

Adversarial Training



$$\min_f \mathbb{E}_{x,y} \max_{|\delta|_\infty \leq \epsilon} \text{loss}(f(x + \delta), y)$$

Works for classification tasks and small perturbation set